

Problem Set 3

Hanyu Li (25346841)

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Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in .pdf form.
- This problem set is due before 23:59 on Wednesday March 25, 2026. No late assignments will be accepted.

Question 1

We are interested in how governments' management of public resources impacts economic prosperity. Our data come from Alvarez, Cheibub, Limongi, and Przeworski (1996) and is labelled `gdpChange.csv` on GitHub. The dataset covers 135 countries observed between 1950 or the year of independence or the first year for which data on economic growth are available ("entry year"), and 1990 or the last year for which data on economic growth are available ("exit year"). The unit of analysis is a particular country during a particular year, for a total $> 3,500$ observations.

- Response variable:
 - `GDPWdiff`: Difference in GDP between year t and $t-1$. Possible categories include: "positive", "negative", or "no change"
- Explanatory variables:
 - `REG`: 1=Democracy; 0=Non-Democracy
 - `OIL`: 1=if the average ratio of fuel exports to total exports in 1984-86 exceeded 50%; 0= otherwise

Please answer the following questions:

1. Construct and interpret an unordered multinomial logit with `GDPWdiff` as the output and "no change" as the reference category, including the estimated cutoff points and coefficients.

```
1 # clean data
2 gdp_clean <- gdp_data |>
3   select(GDPWdiff, REG, OIL) |>
4   filter(!is.na(GDPWdiff) & !is.na(REG) & !is.na(OIL)) |>
5   mutate(
6     GDPWdiff_t = case_when(
7       GDPWdiff > 0 ~ "positive",
8       GDPWdiff < 0 ~ "negative",
9       TRUE ~ "no change"
10    ),
11    GDPWdiff_unord = relevel(as.factor(GDPWdiff_t), ref = "no change"),
12    GDPWdiff_ord = factor(GDPWdiff_t,
13                          levels = c("negative", "no change", "positive")
14    ),
15    ordered = TRUE)
16 head(gdp_clean)
17
18 # 1.1 construct an unordered multinomial logit model
19 unord_ml <- multinom(GDPWdiff_unord ~ REG + OIL, data = gdp_clean)
20
21 # show raw coefficients
22 coef(unord_ml)
23
24 # calculate OR by exponentiating the coefficients
25 exp(coef(unord_ml))
```

Based on the research background, we have our unordered multinomial logit regression as follows:

$$\ln \left(\frac{P(Y_i = \text{negative})}{P(Y_i = \text{no change})} \right) = \beta_{0,\text{negative}} + \beta_{1,\text{negative}} \text{REG}_i + \beta_{2,\text{negative}} \text{OIL}_i$$
$$\ln \left(\frac{P(Y_i = \text{positive})}{P(Y_i = \text{no change})} \right) = \beta_{0,\text{positive}} + \beta_{1,\text{positive}} \text{REG}_i + \beta_{2,\text{positive}} \text{OIL}_i$$

And we have the regression outcome table:

- Here, the two intercepts represent: for a non-democratic country (`REG` = 0) whose average ratio of fuel exports to total exports doesn't exceed 50% (`OIL` = 0), the odds of having positive GDP growth versus no change are expected to be 93.11, while the odds of seeing negative GDP growth versus no change are 44.94.

Table 1: Unordered Multinomial Logit Regression (Odds Ratios)

	<i>Dependent variable:</i>	
	negative	positive
	(1)	(2)
REG	3.972*** (0.769)	5.865*** (0.767)
OIL	119.578*** (6.885)	97.156*** (6.885)
Constant	44.942*** (0.271)	93.108*** (0.269)
Akaike Inf. Crit.	4,690.770	4,690.770
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

- Holding the fuel export ratio (OIL) constant, the odds of seeing negative GDP growth versus no change multiply by a factor of 3.972 when a country is a democracy (REG = 1) compared to a non-democracy (REG = 0) and the odds of observing positive GDP growth versus no change multiply by a factor of 5.865 when a country is a democracy compared to a non-democracy.
- Holding regime (REG) constant, the odds of seeing negative GDP growth versus no change multiply by a factor of 119.578 when a country's average ratio of fuel exports to total exports exceeds 50% (OIL = 1) compared to when this ratio is 50% or less (OIL = 0). Similarly, the odds of observing positive GDP growth versus no change multiply by a factor of 97.156 when a country's fuel export ratio surpasses 50% compared to when it does not.

To interpret the substantive impact of our predictors using predicted probabilities, we generated predictions for hypothetical cases while isolating the effect of each variable. Codes and outcomes are displayed below:

```

1 # create fake data to show predicted probability
2 fake_data <- data.frame(REG = c(0, 1, 0, 0), OIL = c(0, 0, 0, 1))
3 pred_probs <- predict(unord_ml, newdata = fake_data, type = "probs")
4 cbind(fake_data, pred_probs)

```

REG	OIL	no change	negative	positive
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1	0	0	7.191671e-03	0.3232070	0.6696013
2	1	0	1.378186e-03	0.2460217	0.7526001
3	0	0	7.191671e-03	0.3232070	0.6696013
4	0	1	6.934296e-05	0.3726529	0.6272778

- Holding the fuel export variable constant at $OIL = 0$, we compare a non-democracy ($REG = 0$) to a democracy ($REG = 1$). As the results show, when a country transitions to a democracy, the predicted probability of experiencing negative GDP growth decreases from approximately 32.3% to 24.6%. Conversely, the predicted probability of experiencing positive GDP growth increases from 67.0% to 75.3%.
 - To evaluate the substantive effect of fuel exports, we hold regime type constant at non-democracy ($REG = 0$). When a country's average ratio of fuel exports to total exports exceeds 50% ($OIL = 1$) compared to when it does not ($OIL = 0$), the predicted probability of experiencing negative GDP growth increases from approximately 32.3% to 37.3%. Concurrently, the predicted probability of positive GDP growth decreases from 67.0% to 62.7%.
2. Construct and interpret an ordered multinomial logit with `GDPWdiff` as the outcome variable, including the estimated cutoff points and coefficients.

```

1 # 1.2 construct an ordered multinomial logit model
2 # run ordered multinomial logit model
3 ord_ml <- polr(GDPWdiff_ord ~ REG + OIL, data = gdp_clean, Hess = TRUE)
4
5 # show coefficients, standard errors, and cutoffs
6 summary(ord_ml)
7
8 # extract the estimated cutoff points
9 ord_ml$zeta
10
11 # calculate Odds Ratios (OR) by exponentiating the coefficients
12 exp(cbind(OR = coef(ord_ml), confint(ord_ml)))

```

Based on the ordinal nature of our dependent variable (**negative - no change - positive**), our ordered logit regression function is specified as:

$$\ln \left(\frac{P(Y_i \leq j)}{1 - P(Y_i \leq j)} \right) = \mu_j - (\beta_1 REG_i + \beta_2 OIL_i)$$

Where $j \in \{\text{negative, no change}\}$, and μ_j represents the estimated cutoff points on the latent scale.

And we have the regression outcome table:

Table 2: Ordered Multinomial Logit Regression(Odds Ratios)

	<i>Dependent variable:</i>
	GDPWdiff_ord
REG	1.490*** (0.075)
OIL	0.820*** (0.116)
μ_1 (negative — no change)	-0.731
μ_2 (no change — positive)	-0.710
Observations	3,721
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

From the regression outcomes,we can conclude that:

- The model estimates two latent cutoff points ($\mu_1 = -0.731$ for negative — no change, and $\mu_2 = -0.710$ for no change — positive). If a country's underlying latent economic score is below -0.731 , the model predicts negative GDP growth; if the score falls between -0.731 and -0.710 , it predicts no change; and if the score is above -0.710 , it predicts positive GDP growth.
- Holding the fuel export ratio constant, for a democratic country,the odds of having a higher degree of GDP growth are 1.490times higher compared to a non-democratic country.
- Holding regime constant, the odds of moving to a higher GDP growth category multiply by a factor of 0.820 when a country's average ratio of fuel exports to total exports exceeds 50% compared to when this ratio is 50% or less.

Whats more,we still generated fake data to predict probability and calculated probability changes:

```

1 # use fake data and generate predicted probabilities
2 fake_data_ord <- data.frame(REG = c(0, 1, 0, 0), OIL = c(0, 0, 0, 1))
3 pred_probs_ord <- predict(ord_ml, newdata = fake_data_ord, type = "probs"
4 )
5 cbind(fake_data_ord, pred_probs_ord)

```

```

5
6 # calculate probability changes
7 diff_REG <- pred_probs_ord[2, ] - pred_probs_ord[1, ]
8 diff_OIL <- pred_probs_ord[4, ] - pred_probs_ord[3, ]
9 rbind("Effect of REG (0 to 1)" = diff_REG,
10      "Effect of OIL (0 to 1)" = diff_OIL)

```

	REG	OIL	negative	no change	positive
1	0	0	0.3249362	0.004555476	0.6705083
2	1	0	0.2442235	0.003839727	0.7519368
3	0	0	0.3249362	0.004555476	0.6705083
4	0	1	0.3699431	0.004836151	0.6252207

Table 3: Changes in Predicted Probabilities

<i>Change in Variable</i>	<i>Predicted Outcomes</i>		
	negative	no change	positive
Effect of REG (0 → 1)	−8.07	−0.07	+8.14
Effect of OIL (0 → 1)	+4.50	+0.03	−4.53

- Holding the fuel export ratio constant, transitioning from a non-democracy to a democracy decreases the probability of experiencing negative GDP growth by 8.07 percentage points and increases the probability of positive growth by 8.14 percentage points.
- Holding regime type constant, when a country's average ratio of fuel exports to total exports exceeds 50%, the probability of negative GDP growth increases by 4.50 percentage points, while the probability of positive growth decreases by 4.53 percentage points.

Question 2

Consider the data set `MexicoMuniData.csv`, which includes municipal-level information from Mexico. The outcome of interest is the number of times the winning PAN presidential candidate in 2006 (`PAN.visits.06`) visited a district leading up to the 2009 federal elections, which is a count. Our main predictor of interest is whether the district was highly contested, or whether it was not (the PAN or their opponents have electoral security) in the previous federal elections during 2000 (`competitive.district`), which is binary (1=close/swing district, 0="safe seat"). We also include `marginality.06` (a measure of poverty) and `PAN.governor.06` (a dummy for whether the state has a PAN-affiliated governor) as additional control variables.

- (a) Run a Poisson regression because the outcome is a count variable. Is there evidence that PAN presidential candidates visit swing districts more? Provide a test statistic and p-value.

```

1 # load data
2 mexico_elections <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/
  StatsII-2026/main/datasets/MexicoMuniData.csv")
3
4 # run poisson regression model
5 poi_ml <- glm(PAN.visits.06 ~ competitive.district + marginality.06 + PAN
  .governor.06,
6               data = mexico_elections, family = poisson)
7
8 # show coefficients, standard errors, and z-values
9 summary(poi_ml)
10
11 # calculate Odds Ratios by exponentiating the coefficients
12 exp(cbind(OR = coef(poi_ml), confint(poi_ml)))

```

Based on the research background and count data nature, we specified the Poisson regression model as:

$$\ln(\hat{\lambda}_i) = \beta_0 + \beta_1 \text{competitive.district}_i + \beta_2 \text{marginality.06}_i + \beta_3 \text{PAN.governor.06}_i$$

Where $\hat{\lambda}_i$ represents the expected mean number of visits.

To test whether candidates visit swing districts more, we establish the following hypothesis for the coefficient of `competitive.district` (β_1):

- **Null Hypothesis (H_0):** $\beta_1 = 0$. District competitiveness has no effect on the expected number of visits.
- **Alternative Hypothesis (H_a):** $\beta_1 \neq 0$. District competitiveness has an effect on the expected number of visits.

And we have the regression outcome table (with exponentiated coefficients):

Coefficients:

Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-3.81023	0.22209	-17.156	<2e-16 ***
competitive.district	-0.08135	0.17069	-0.477	0.6336
marginality.06	-2.08014	0.11734	-17.728	<2e-16 ***
PAN.governor.06	-0.31158	0.16673	-1.869	0.0617 .

Table 4: Poisson Regression (Odds Ratios)

	<i>Dependent variable:</i>
	PAN.visits.06
competitive.district	0.922 (0.171)
marginality.06	0.125*** (0.117)
PAN.governor.06	0.732* (0.167)
Constant	0.022*** (0.222)
Observations	2,407
Log Likelihood	−645.606
Akaike Inf. Crit.	1,299.213
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Based on the model summary, the test statistic (z-value) for `competitive.district` is -0.477 , and the corresponding p-value is 0.6336 . Because the p-value (0.634) is greater than the threshold 0.05 significance level, we don't have enough evidence to reject the null hypothesis. This means that whether a district was a swing district in the previous election isn't statistically associated with the increase of the expected number of visits from the winning PAN presidential candidate.

(b) Interpret the `marginality.06` and `PAN.governor.06` coefficients.

```
1 # check equal variance assumption (over-dispersion)
2 dispersiontest(poi_ml)
```

```
z = 1.0668, p-value = 0.143
alternative hypothesis: true dispersion is greater than 1
sample estimates:
dispersion
2.09834
```

Before interpreting the coefficients, we conducted a dispersion test to check the equal variance assumption of the Poisson model. The test yields a p-value of 0.143 . Because $0.143 > 0.05$, we fail to reject the null hypothesis that the true dispersion is not greater than 1. Therefore, the Poisson model is correctly specified and has valid explanatory power.

Therefore, we can interpret coefficients in the model as follows:

- **`marginality.06`:** Holding other variables constant, a one-unit increase in the marginality (poverty) index has a multiplicative effect on the expected mean of visits by 0.125 ($e^{-2.080}$). This means that for every one-unit increase in the marginality index, we expect to see 87.5% fewer visits from the winning PAN presidential candidate.
- **`PAN.governor.06`:** Holding other variables constant, having a PAN-affiliated governor has a multiplicative effect on the expected mean of visits by 0.732 ($e^{-0.312}$). This implies that states with a PAN governor expect to see approximately 26.8% fewer visits compared to states without a PAN governor.

(c) Provide the estimated mean number of visits from the winning PAN presidential candidate for a hypothetical district that was competitive (`competitive.district=1`), had an average poverty level (`marginality.06 = 0`), and a PAN governor (`PAN.governor.06=1`).

To answer this, we utilize our model to generate a fitted value for this hypothetical scenario:

```
1 # use fake data and generate predicted expected counts
2 pred_data <- data.frame(competitive.district = 1, marginality.06 = 0, PAN
  .governor.06 = 1)
3 pred_count <- predict(poi_ml, newdata = pred_data, type = "response")
4 cbind(pred_data, Predicted_Visits = pred_count)
```

competitive.district	marginality.06	PAN.governor.06	Predicted_Visits
1	0	1	0.01494818

- Based on the prediction, for a competitive district (`competitive.district = 1`) with an average poverty level (`marginality.06 = 0`) and a PAN governor (`PAN.governor.06 = 1`), the estimated mean number of visits is $\hat{\lambda} \approx 0.015$. This suggests that a district with these characteristics would barely expect any visits at all from the winning candidate.